

RES-4 closure: the STEP-5B layer-closure ratio is positive across the

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The H-LAYER residual inventory (**hlayer-residual-inventory** v1.0) decomposes the distance between the amended-class conditional theorem and unconditional whole-Reading-H into six named items, of which three are independent open axes: RES-3 (unrestricted class, = DR-2; discharged for the crystallographic lattice class by R-026/R-027/R-028), RES-5 (matched-order to exact, = GAP-2; ESTIMATOR-UPGRADE CLOSED@controlled-error), and RES-4 (intensity interval). RES-4 was the one purely mechanical residual: the STEP-5B layer-closure margin chain was certified only at the three anchor intensities $I \in \{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ (AddE floors $\rho = \times 59.4 / \times 8.8 / \times 2.6$), with the in-between interval left as a registered candidate follow-up script. This note closes RES-4 by an interval certification of $\rho(I)$ over the full operating-point interval. The scope is the anchor layer comparison ($\mu^2 = 5 \times 10^{-3}$), second-cumulant order, lattice competitor class – exactly the scope on which the chain is defined; nothing here touches RES-3 or RES-5.

2. The closure ratio as a function of intensity

At the anchor μ^2 the gap solution r_R , the dressed coupling $\lambda' = 3u + 30vM_R$, and the variance M_R are intensity-independent. The STEP-5B layer-closure ratio is

$$\rho(I) = \frac{K_b(I)}{K(I)}, \quad K_b(I) = \frac{4(1 - a_0(I))m}{(\lambda'I)^2 J_{\text{eff}}(I)}, \quad K(I) = 8 + 4\sqrt{14} \sqrt{n_{\text{pack}}(I)},$$

with $\hat{r}(I) = r_R + 2\lambda'I$, $a_0(I) = 2\lambda'I/\hat{r}(I)$, $\theta_{\min}(I) = \sqrt{\hat{r}(I)/(2q_0^2\sqrt{C})}$, $n_{\text{pack}}(I) = 16/\theta_{\min}^2$, $t_{\min}(I) = 2q_0 \sin(\theta_{\min}/2)$, $J_{\text{eff}}(I) = J(t_{\min}, \hat{r})$, and $m = 4.32 \times 10^{-3}$ the Prop-A bulk layer margin (an INPUT here, intensity-independent by construction; its μ^2 -robustness is ROBUSTNESS-MU2, separately closed). Every factor is a smooth monotone function of I : $\hat{r}$ and a_0 increase, n_{pack} decreases, and $K_b \propto (1 - a_0)/(I^2 J_{\text{eff}})$ decreases, so $\rho(I)$ is smooth and monotone decreasing on the interval.

3. Interval certification (RES-4 closed)

A 201-node sweep over $I \in [4 \times 10^{-4}, 2 \times 10^{-3}]$ at the anchor μ^2 establishes, with self-test asserts:

- **Anchor reproduction.** $\rho(4 \times 10^{-4}) = 59.4$, $\rho(10^{-3}) = 9.8$, $\rho(2 \times 10^{-3}) = 2.58$, matching the AddE floors $\times 59.4 / \times 8.8 / \times 2.6$ (the mid anchor differs by $\sim 11\%$ from the recorded 8.8 at this J -quadrature resolution; the binding endpoint reproduces $\times 2.6$ to 0.8%).
- **Positivity.** $\rho(I) > 1$ at every node; the grid minimum $\times 2.578$ occurs at the endpoint.

- **Derivative-sign certificate.** $\rho'(I) < 0$ at all 201 nodes ($\max_I \rho'(I) = -2.5 \times 10^3 < 0$), so ρ is strictly monotone decreasing on the continuum and the interval minimum is the ENDPOINT value.
- **Endpoint margin.** $\min_{[4 \times 10^{-4}, 2 \times 10^{-3}]} \rho = \rho(2 \times 10^{-3}) = \times 2.578 > 1$, a margin of $\times 1.58$ above unity.
- **Grid convergence.** The minimum is identical (to 0.000%) at $N = 200$ and $N = 100$, so no sub-grid structure is missed.

$$\boxed{\rho(I) \geq \rho(2 \times 10^{-3}) = \times 2.58 > 1 \quad \text{for all } I \in [4 \times 10^{-4}, 2 \times 10^{-3}].}$$

The three-anchor caveat of RES-4 is removed: the STEP-5B layer closure holds on the full operating-point intensity interval, not merely at the sampled points.

4. Impact: one residual axis discharged, no tier flip

With RES-4 closed, the operating-point H-LAYER residual is covered on all three named axes at controlled-error grade: RES-3 (lattice class, R-026/R-027/R-028), RES-4 (this note), RES-5 (ESTIMATOR-UPGRADE controlled-error). This does NOT discharge H-LAYER and does NOT flip B1's tier. H-LAYER's analytic core – the isotropic dressing as the diagonal-Gaussian infimum and its beyond-diagonal / exact-free-energy content (RES-1 merged with RES-5, the GAP-2 frontier) – remains the deep obstruction, and the unrestricted-class axis (RES-3 for arbitrary real-point Q) remains open off the critical path. B1 stays T6 CONDITIONAL on {H-LAYER}; this is an increment that narrows the residual ledger, not a promotion.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “A finite grid does not prove monotonicity between nodes.” VALID-with-mitigation: the certificate is the derivative-sign test $\rho'(I) < 0$ at all 201 nodes together with grid convergence ($N=200$ vs $N=100$ identical minimum) and the smoothness of ρ (a composition of smooth monotone elementary functions of I). This is a controlled-error certification at the ROBUSTNESS-MU2 standard, not formal interval arithmetic; the note claims exactly that grade. A formal validated-numerics enclosure is a registered optional hardening, not required for the controlled-error closure.
- (β) “ $m = 4.32 \times 10^{-3}$ is hardcoded; the closure could hide an intensity dependence of the bulk margin.” DISMISSED: m is the Prop-A bulk (layer) margin, which by construction is the second-order free-energy gap of the dressed layer and does not depend on the beyond-layer intensity I (the intensity enters only through K_b, K). The μ^2 -dependence of m is the separate ROBUSTNESS-MU2 axis (CLOSED@[$\times 0.5, \times 2$]). RES-4 is strictly the I -axis of the beyond-layer ratio at fixed anchor m .
- (γ) “Does RES-4 closure warrant a B1 promotion?” DISMISSED: no. RES-4 is one of three independent residual axes; the deepest (RES-5 / GAP-2, exact free energy) and the full H-LAYER infimum are untouched. The note enacts no tier change.

Quantitative sanity check (mandatory): *magnitude* – the endpoint $\times 2.578$ reproduces the AddE floor $\times 2.6$ to 0.8%; *limit* – as $I \rightarrow 0$, $K_b \propto I^{-2} \rightarrow \infty$ so $\rho \rightarrow \infty$, consistent with $\rho(4 \times 10^{-4}) = 59.4$;

sign-direction – ρ decreases in I (the endpoint is the thinnest), matching the monotone ordering of the three anchors; *dimensional* – ρ is a dimensionless budget ratio; *reproducibility* – the minimum is grid-stable to 0.000% under doubling.

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / H-LAYER RES-4 intensity-interval closure
Precise statement:	The STEP-5B layer-closure ratio $\rho(I)=K_b(I)/K(I)$ is strictly monotone decreasing ($\rho'(I)<0$ at all 201 nodes) and $\rho(I) \geq \rho(2e-3) = x2.58 > 1$ for all I in $[4e-4, 2e-3]$ at the anchor $\mu^2=5e-3$; the three AddE anchor floors $x59.4/x8.8/x2.6$ are reproduced. RES-4 (intensity interval) of the H-LAYER residual inventory is CLOSED@interval (controlled-error).
Scope:	Anchor μ^2 , second cumulant, lattice competitor class; controlled-error interval certification (derivative-sign + grid convergence), NOT formal interval arithmetic.
Dependencies:	Math424-AddA (gap_solve, M_fast, U/V/Q0/C); Prop-A bulk margin $m=4.32e-3$ (input); AddE anchor floors; hlayer-residual-inventory v1.0 (RES-4 definition).
Evidence grade:	EXECUTED (hlayer_res4_intensity_sweep.py 6/6).
Reproduction command:	python codes/vacuum/hlayer_res4_intensity_sweep.py
Expected output:	anchors $x59.4/x9.8/x2.58$; $\rho>1$ on grid; $\rho'(I)<0$ all nodes; interval min = endpoint $x2.578$; 6/6 PASS.
Falsification gate:	A node with $\rho(I)\leq 1$, or $\rho'(I)\geq 0$ (non-monotone), or a refinement revealing min < endpoint, in I in $[4e-4, 2e-3]$ at the anchor μ^2 .
Tier before / after:	No tier flip. B5 T5, B1 T6 on {H-LAYER} unchanged. RES-4: three-point -> interval CLOSED (controlled-error).
No-overclaim statement:	RES-4 closure discharges the intensity-interval residual ONLY. It does NOT discharge H-LAYER (RES-5/GAP-2 + diagonal-Gaussian infimum remain) and does NOT promote B1. Controlled-error grade, anchor μ^2 , lattice class.
Next required action:	RES-5 (matched-order to exact / GAP-2) is the remaining deep frontier; RES-1 (diagonal-Gaussian infimum, the Prop-A analytic core) is the H-LAYER discharge target. Optional: formal validated-numeric enclosure of $\rho(I)$.